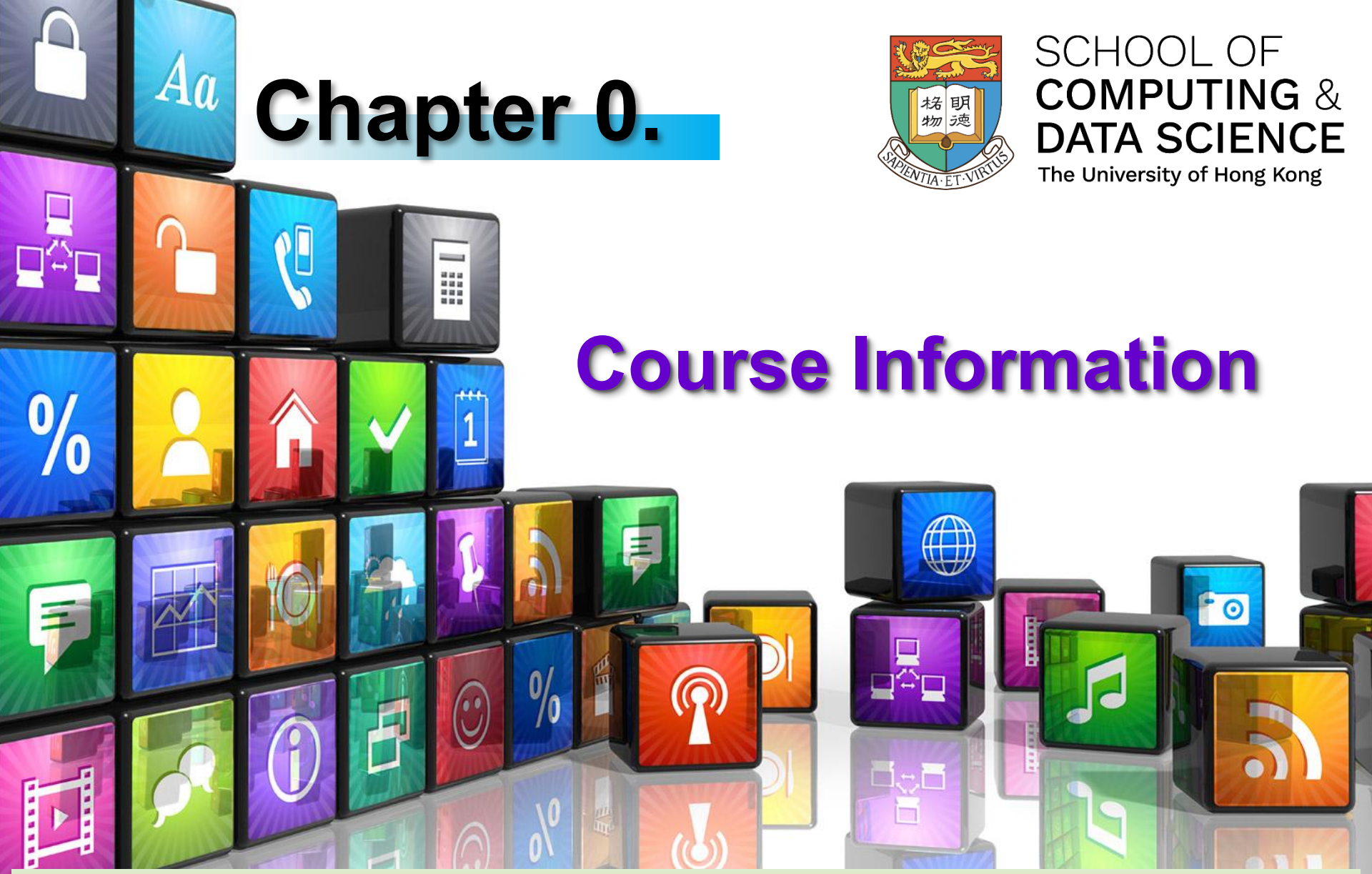
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SCHOOL OF COMPUTING & DATA SCIENCE
The University of Hong Kong



Course Information

2026-2027
COMP3330 Interactive Mobile Application Design and Programming
Dr. T.W. Chim (E-mail: twchim.cs@hku.hk)

About us

Instructor:

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Pre-requisite

- COMP2396 / CSIS0396 Object-oriented Programming and Java
  - Because Java is one of the basic languages for Android apps development.
  - Android Studio also supports Kotlin but Kotlin is just a variation of Java and is intended to simplify the use of Java.
- How about C and C++?
  - From Android Studio 2.2 on, you can add C and C++ code to your app by compiling it into a native library that Gradle can package with your APK (default build tool for native libraries is CMake).
  - Your Java / Kotlin code can then call functions in your native library through the Java Native Interface (JNI).
  - Reference: https://developer.android.com/studio/projects/add-native-code.html

Course Learning Outcomes

- **CLO 1. Understand the basics about mobile apps development, including features of mobile devices, types and lifecycle of applications and markets**

- **CLO 2. Be able to develop mobile applications using common IDEs in the community**



Course Learning Outcomes

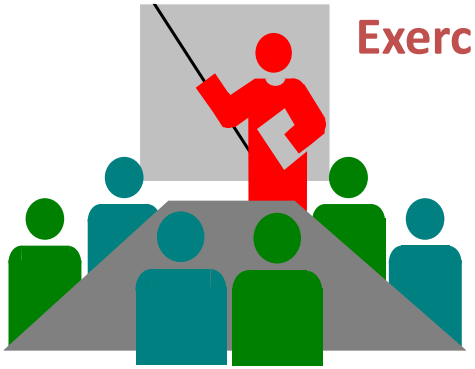
- **CLO 3. Understand various components for interactive mobile application development including user interface design, graphics, handling of concurrency, local database, network, multimedia, use of sensors and location services, testing and publication of mobile apps**
- **CLO 4. Exercise project management skills including ideation, reading, writing, coding, and presentation**
- **CLO 5. Possess the mentality to deal with the fast-changing future mobile application development world**

Course outline

- Fundamentals of mobile application development
  - Basic principles and constraints of mobile apps [CLO 1]
  - Lifecycle of mobile apps [CLO 1]
  - Design patterns of mobile apps development [CLO 2]
  - Introduction of modern cross-platform tools [CLO 5]
  - Testing and publication of mobile applications [CLO 1]
- Technical components of mobile application development
  - User interface [CLO 1, 2, 3]
  - Graphics and multimedia [CLO 1, 2, 3]
  - Handling of concurrency in mobile apps [CLO 1, 2, 3]
  - Local database and network [CLO 1, 2, 3]
  - Use of sensors and location services [CLO 1, 2, 3]
  - Server supported mobile applications [CLO 1, 2, 3]
  - Advanced topics [CLO 1, 2, 3]
- Integration of knowledge and project work
  - Project ideation, writing, implementation, management, and presentation [CLO 1, 2, 3, 4, 5]

Course Structure

Lectures, Tutorials, Exercises



Final Exam (Written)



- Tutorial Exercises
- 1 Group Project



Consultation & Moodle Discussion Forums (recommended)



Assessment

- Continuous Assessment [70%]
  - Tutorial Exercises [20%]
  - Group Project (Group size: 1 – 5 students) [50%]
    - App Proposal (max 5 pages) [10%]
    - Final App Presentation & Demonstration [5%]
    - App Trailer [10%]
    - Workable App Product [5%]
    - Report [10%]
    - Peer Learning Notes showing how you used AI in your development (e.g., graphics generation) [10%]

- Final Examination [30%]
  - Written
  - Can bring into the exam venue an A4-sized cheat sheet with notes written or printed on both sides

COMP3330 Interactive Mobile Application Design and Programming



This is a certified Communication-intensive (CI) Course which meets all of the requirements endorsed by HKU's Senate, including

- the teaching and assessment of oral and written communication 'literacies'; and
- at least 40% of the course grade assigned to communication-rich assessment tasks.

What communication knowledge and skills will students learn in this course?

- In this course, students will learn how to effectively communicate app ideas and introduce the developed app and its features in both oral and written forms.
- Oral Literacy:
  - Students will be taught how to deliver an effective presentation. They will develop and reinforce oral communication skills (in particular being able to present ideas clearly and concisely, being open to different perspectives, being able to engage in discussion actively, showing confidence in providing constructive criticism to peers and responding to constructive criticism from others, and being able to participate in teamwork) and in conducting interactive presentations and engaging discussions, and in delivering effective presentation slides.
- Written Literacy:
  - Students will learn how to write a technical report that is precise and concise with an appropriate tone. They will analyze exemplars of past reports for successful features of technical report writing. Students will need to explain the technical content to the general public.

What does a good communicator look like in this course?

- Be able to effectively articulate and present their ideas and points of view
- Be open to different perspectives
- Be able to engage in a discussion in a professional and scholarly manner
- Be able to show confidence in providing constructive criticism to peers and respond to constructive criticism from peers and teachers
- Be able to demonstrate leadership and collaboration as team players

Meeting Time

- Lecture:
  - Every Monday, 9:00am – 10:50am @ CYPP3
- Tutorial:
  - Every Thursday, 12:00pm – 12:50pm @ CYPP3
  - Except Sep 3

- Some classes will be reserved for group project presentations.

Schedule

The teaching plan maybe rescheduled according to the actual progress of the course. Also some lectures (tutorials) may be changed into tutorials (lectures) with prior notice.

	Week	Date (Mon)	Activity (09:00 – 10:50)	Date (Thu)	Activity (12:00 – 12:50)
Semester 1	1			Sep 3	Lecture
	2	Sep 7	Lecture	Sep 10	Tutorial 1
	3	Sep 14	Lecture	Sep 17	Tutorial 2
	4	Sep 21	Lecture	Sep 24	Tutorial 3
	5	Sep 28	Lecture	Oct 1	National Day
	6	Oct 5	Lecture	Oct 8	Tutorial 4
	7	Oct 12	Reading Week	Oct 15	Reading Week
	8	Oct 19	Chung Yeung Festival	Oct 22	Lecture
	9	Oct 26	Lecture	Oct 29	Tutorial 5
	10	Nov 2	Lecture	Nov 5	Tutorial 6
	11	Nov 9	Lecture	Nov 12	Tutorial 7
	12	Nov 16	Lecture	Nov 19	Tutorial 8
	13	Nov 23	Final App Presentation	Nov 26	Final App Presentation
	14	Nov 30	Final App Presentation		

Lecture: 20 hours + Tutorials: 8 hours + Presentation: 5 hours

Reference

- Documentation
  - http://developer.android.com/

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End

